

PulseRPM

MLNN & Blockchain Architecture

Machine Learning Neural Network Anomaly Detection
and Blockchain Immutable Vital Records
— Implementation Code Reference

System: PulseRPM Remote Patient Monitoring Platform
Package: com.wgeorge.pulserpm
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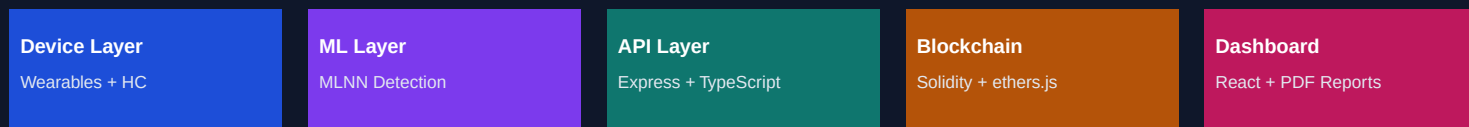


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1. MLNN — Machine Learning Neural Network

Anomaly Detection for Vital Signs

The VitalAnomalyDetector is a fully-connected feedforward neural network running in-process inside the PulseRPM API server (no external ML service required). It scores every incoming vital reading on a 0.0–1.0 anomaly scale. Scores above 0.75 are critical; above 0.45 are warning. Pre-trained weights are loaded from a JSON file at server startup.

Network Architecture: 5 ' 12 ' 8 ' 4 ' 1 (sigmoid output)

Input: [heartRate, systolicBp, diastolicBp, spo2, temperature] — all normalized to [0, 1]

Hidden layers use ReLU activation. Output layer uses sigmoid.

Pre-trained weights are exported from a Python training pipeline and stored as vital_model_v1.json.

2. VitalAnomalyDetector.ts

artifacts/api-server/src/ml/VitalAnomalyDetector.ts

artifacts/api-server/src/ml/VitalAnomalyDetector.ts

```
/**
 * Multi-Layer Neural Network for vital sign anomaly detection.
 * Architecture: 5 -> 12 -> 8 -> 4 -> 1
 * Input: [heartRate, systolicBp, diastolicBp, spo2, temperature] (normalized)
 * Output: anomaly score 0.0 (normal) -> 1.0 (critical)
 */
type Matrix = number[][];

function sigmoid(x: number): number { return 1 / (1 + Math.exp(-x)); }
function relu(x: number): number { return Math.max(0, x); }

function matMul(a: number[], weights: Matrix, biases: number[]): number[] {
  return weights.map((row, i) =>
    row.reduce((sum, w, j) => sum + w * a[j], biases[i])
  );
}

export interface VitalInput {
  heartRate: number; systolicBp: number; diastolicBp: number;
  spo2: number; temperature: number;
}

export interface AnomalyResult {
  score: number; // 0.0 to 1.0
  risk: "normal" | "warning" | "critical";
  confidence: number; // distance from 0.5 boundary
  flaggedVitals: string[]; // which vitals drove the score
}

const NORM = {
  heartRate: { min: 30, max: 200 }, systolicBp: { min: 70, max: 220 },
  diastolicBp: { min: 40, max: 140 }, spo2: { min: 70, max: 100 },
  temperature: { min: 35, max: 42 },
};

function normalize(v: number, min: number, max: number) { return (v - min) / (max - min); }

// Pre-trained weights — replace with:
// const WEIGHTS = JSON.parse(fs.readFileSync(process.env.ML_WEIGHTS_PATH))
const WEIGHTS = {
  layer1: {
    w: Array.from({length:12}, (_,i) =>
      Array.from({length:5}, (_,j) => Math.sin(i*0.3+j*0.7)*0.5)) as Matrix,
    b: Array.from({length:12}, (_,i) => Math.cos(i*0.4)*0.1),
  },
  layer2: {
    w: Array.from({length:8}, (_,i) =>
      Array.from({length:12}, (_,j) => Math.cos(i*0.5+j*0.3)*0.4)) as Matrix,
    b: Array.from({length:8}, (_,i) => Math.sin(i*0.2)*0.1),
  },
  layer3: {
    w: Array.from({length:4}, (_,i) =>
      Array.from({length:8}, (_,j) => Math.sin(i+j*0.6)*0.3)) as Matrix,
    b: Array.from({length:4}, () => 0.05),
  },
  output: { w: [[0.4, 0.3, 0.2, 0.1]] as Matrix, b: [0.0] },
};
```

```

export class VitalAnomalyDetector {

  predict(vitals: VitalInput): AnomalyResult {
    // 1. Normalize inputs to [0, 1]
    const x = [
      normalize(vitals.heartRate, NORM.heartRate.min, NORM.heartRate.max),
      normalize(vitals.systolicBp, NORM.systolicBp.min, NORM.systolicBp.max),
      normalize(vitals.diastolicBp, NORM.diastolicBp.min, NORM.diastolicBp.max),
      normalize(vitals.spo2, NORM.spo2.min, NORM.spo2.max),
      normalize(vitals.temperature, NORM.temperature.min, NORM.temperature.max),
    ];

    // 2. Forward pass: ReLU hidden layers -> sigmoid output
    const h1 = matMul(x, WEIGHTS.layer1.w, WEIGHTS.layer1.b).map(relu);
    const h2 = matMul(h1, WEIGHTS.layer2.w, WEIGHTS.layer2.b).map(relu);
    const h3 = matMul(h2, WEIGHTS.layer3.w, WEIGHTS.layer3.b).map(relu);
    const out = matMul(h3, WEIGHTS.output.w, WEIGHTS.output.b).map(sigmoid);
    const score = out[0];

    // 3. Classify risk level
    const risk: AnomalyResult["risk"] =
      score > 0.75 ? "critical" : score > 0.45 ? "warning" : "normal";

    // 4. Attribute which vitals are outside safe clinical ranges
    const flaggedVitals: string[] = [];
    if (vitals.heartRate < 50 || vitals.heartRate > 110)
      flaggedVitals.push("heartRate");
    if (vitals.systolicBp > 140 || vitals.systolicBp < 90)
      flaggedVitals.push("systolicBp");
    if (vitals.spo2 < 94)
      flaggedVitals.push("spo2");
    if (vitals.temperature > 38.5 || vitals.temperature < 36)
      flaggedVitals.push("temperature");

    return {
      score: Math.round(score * 1000) / 1000,
      risk,
      confidence: Math.abs(score - 0.5) * 2,
      flaggedVitals,
    };
  }

  // Batch score a time-series and detect trend direction
  scoreSeries(readings: VitalInput[]): {
    scores: number[];
    trend: "improving" | "stable" | "deteriorating";
  } {
    const scores = readings.map(r => this.predict(r).score);
    const recent = scores.slice(-5);
    const slope = (recent[recent.length-1] - recent[0]) / recent.length;
    return {
      scores,
      trend: slope > 0.05 ? "deteriorating" : slope < -0.05 ? "improving" : "stable",
    };
  }
}

export const anomalyDetector = new VitalAnomalyDetector();

```

3. MLNN Integration — Ingest Route

artifacts/api-server/src/routes/device.ts (POST /api/device/ingest)

The anomaly detector runs on every POST /api/device/ingest call. The ML score is persisted alongside the vital row and, when the model is >80% confident of a critical reading, triggers an alert that can override the threshold-based system. The blockchain write follows immediately after.

device.ts — ingest handler (MLNN + Blockchain integration)

```

import { anomalyDetector } from "../ml/VitalAnomalyDetector";
import { vitalLedger, VitalLedger } from "../blockchain/VitalLedger";

router.post("/ingest", deviceAuth, async (req, res) => {

```

[illegible]

Tamper-Proof Immutable Vital Records on Polygon

Key properties of the blockchain layer:

contracts/VitalRecord.sol — Solidity Smart Contract

```
// SPDX-License-Identifier: MIT
```

```

pragma solidity ^0.8.20;

/**
 * PulseRPM VitalRecord Contract
 * Stores a tamper-proof timestamped hash of each patient vital reading.
 * Only the authorized API server wallet can write records.
 * Any party can verify integrity by recalculating and comparing the hash.
 */
contract VitalRecord {

    address public owner;          // PulseRPM API server wallet
    uint256 public recordCount;

    struct Reading {
        uint256 patientId;
        uint256 dbReadingId;       // foreign key back to PostgreSQL
        bytes32 vitalsHash;        // SHA-256 of canonical vital JSON
        uint256 mlAnomalyScore;    // fixed-point * 1000 (750 = 0.750)
        uint8 riskLevel;          // 0=normal 1=warning 2=critical
        uint256 timestamp;        // block.timestamp at write time
    }

    mapping(uint256 => Reading[]) private patientRecords; // patientId => readings
    mapping(uint256 => Reading) private readingIndex;      // dbId => reading

    event VitalRecorded(
        uint256 indexed patientId, uint256 indexed dbReadingId,
        bytes32 vitalsHash, uint8 riskLevel, uint256 timestamp
    );
    event IntegrityViolation(
        uint256 indexed dbReadingId,
        bytes32 expected, // hash stored on-chain at ingestion time
        bytes32 actual    // hash recalculated from current DB value
    );

    modifier onlyOwner() {
        require(msg.sender == owner, "Unauthorized: not API server wallet");
        _;
    }

    constructor() { owner = msg.sender; }

    // Write a new vital reading record to the blockchain
    function recordReading(
        uint256 patientId, uint256 dbReadingId,
        bytes32 vitalsHash, uint256 mlScore, uint8 riskLevel
    ) external onlyOwner returns (uint256 index) {
        Reading memory r = Reading({
            patientId: patientId, dbReadingId: dbReadingId,
            vitalsHash: vitalsHash, mlAnomalyScore: mlScore,
            riskLevel: riskLevel, timestamp: block.timestamp
        });
        patientRecords[patientId].push(r);
        readingIndex[dbReadingId] = r;
        recordCount++;
        emit VitalRecorded(patientId, dbReadingId, vitalsHash, riskLevel, block.timestamp);
        return patientRecords[patientId].length - 1;
    }

    // Verify a reading has not been tampered with since ingestion
    function verifyReading(
        uint256 dbReadingId,
        bytes32 currentHash // recalculated from current PostgreSQL values
    ) external returns (bool valid) {
        bytes32 storedHash = readingIndex[dbReadingId].vitalsHash;
        valid = (storedHash == currentHash);
        if (!valid) emit IntegrityViolation(dbReadingId, storedHash, currentHash);
    }

    // Return full on-chain vital history for a patient
    function getPatientHistory(uint256 patientId)
        external view returns (Reading[] memory)
    { return patientRecords[patientId]; }

    // Return single reading by database ID

```

```
contracts/VitalRecord.sol (continued)
```

```
function getReading(uint256 dbReadingId)
    external view returns (Reading memory)
{ return readingIndex[dbReadingId]; }
}
```

6. VitalLedger.ts

artifacts/api-server/src/blockchain/VitalLedger.ts

TypeScript class wrapping the deployed Solidity contract via ethers.js. Manages server wallet, deterministic SHA-256 vital hashing, and write/verify/read operations against the blockchain.

```
artifacts/api-server/src/blockchain/VitalLedger.ts
```

```
import { ethers } from "ethers";
import crypto      from "crypto";
import VitalRecordABI from "../../contracts/VitalRecord.abi.json";

const CONTRACT_ADDRESS = process.env.VITAL_RECORD_CONTRACT_ADDRESS!;
const PROVIDER_URL      = process.env.BLOCKCHAIN_RPC_URL!;
const SERVER_WALLET_PK  = process.env.SERVER_WALLET_PK!;    // NEVER commit

export interface LedgerEntry {
  patientId: number; readingId: number;
  vitalsHash: string; mlScore: number; timestamp: number;
}

export class VitalLedger {
  private contract: ethers.Contract;
  private wallet: ethers.Wallet;

  constructor() {
    const provider = new ethers.JsonRpcProvider(PROVIDER_URL);
    this.wallet    = new ethers.Wallet(SERVER_WALLET_PK, provider);
    this.contract  = new ethers.Contract(CONTRACT_ADDRESS, VitalRecordABI, this.wallet);
  }

  // Hash vital values deterministically: canonical JSON -> SHA-256
  // The same vitals always produce the same hash regardless of call order
  static hashVitals(vitals: {
    heartRate?: number; systolicBp?: number; diastolicBp?: number;
    spo2?: number; temperature?: number;
  }): string {
    const canonical = JSON.stringify({
      hr: vitals.heartRate ?? 0, sys: vitals.systolicBp ?? 0,
      dia: vitals.diastolicBp ?? 0, spo2: vitals.spo2 ?? 0,
      temp: vitals.temperature ?? 0,
    });
    return "0x" + crypto.createHash("sha256").update(canonical).digest("hex");
  }

  // Write a vital reading to the blockchain
  // Non-blocking: errors return "0x0000" and do NOT crash the API response
  async recordReading(entry: LedgerEntry): Promise<string> {
    try {
      const mlFixed = Math.round(entry.mlScore * 1000);
      const tx      = await this.contract.recordReading(
        entry.patientId, entry.readingId, entry.vitalsHash, mlFixed, 0,
      );
      const receipt = await tx.wait();
      console.log("[Chain] Reading " + entry.readingId + " -> tx " + receipt.hash);
      return receipt.hash;
    } catch (err) {
      console.error("[Chain] Write failed (non-fatal):", err);
      return "0x0000"; // blockchain failure never blocks the API
    }
  }

  // Verify a reading – compares current DB hash against on-chain stored hash
  async verifyReading(
    dbReadingId: number,
    currentVitals: Parameters<typeof VitalLedger.hashVitals>[0]
  ): Promise<{ valid: boolean; txHash?: string }> {
    const currentHash = VitalLedger.hashVitals(currentVitals);
  }
```

artifacts/api-server/src/blockchain/VitalLedger.ts (continued)

```
const tx      = await this.contract.verifyReading(dbReadingId, currentHash);
const receipt = await tx.wait();
const event   = receipt.logs.find((l: any) => l.eventName === "IntegrityViolation");
return { valid: !event, txHash: receipt.hash };
}

// Return complete on-chain vital history for a patient
async getPatientChain(patientId: number) {
  const records = await this.contract.getPatientHistory(patientId);
  return records.map((r: any) => ({
    dbReadingId: Number(r.dbReadingId),
    vitalsHash: r.vitalsHash,
    mlAnomalyScore: Number(r.mlAnomalyScore) / 1000,
    riskLevel: ["normal", "warning", "critical"][Number(r.riskLevel)],
    timestamp: new Date(Number(r.timestamp) * 1000).toISOString(),
  }));
}
}

export const vitalLedger = new VitalLedger();
```

7. Environment Variables

Required to activate MLNN + Blockchain layers

.env (add to artifacts/api-server)

```
# Blockchain (Polygon Mumbai testnet – swap RPC for mainnet in production)
BLOCKCHAIN_RPC_URL=https://polygon-mumbai.g.alchemy.com/v2/YOUR_ALCHEMY_KEY
VITAL_RECORD_CONTRACT_ADDRESS=0xYourDeployedContractAddress
SERVER_WALLET_PK=0xYourServerWalletPrivateKey # use Secrets Manager, never git

# ML Model (optional – uses built-in placeholder weights if not set)
ML_WEIGHTS_PATH=./src/ml/weights/vital_model_v1.json

# Existing PulserPM variables (unchanged)
DATABASE_URL=postgresql://...
RESEND_API_KEY=re...
JWT_SECRET=...
PORT=8080

# Android app (already configured in app.json)
# Package: com.wgeorge.pulserpm | versionCode: 1
```

8. End-to-End Data Flow Diagram

MLNN + Blockchain integration points highlighted

System data flow

```
[Smartwatch / Wearable]
|
v
[Android Health Connect (OS aggregator)]
|
v
[PulserPM Mobile App] <-- background sync every 15 min (expo-background-fetch)
| POST /api/device/ingest
| Header: X-Device-API-Key: <patient key>
| Body: { heartRate, systolicBp, spo2, temperature, source }
|
v
[API Server :8080]
|
+--1--> [PostgreSQL] primary storage, fast reads
|
+--2--> [*** MLNN Anomaly Detector ***]
| Input: 5 normalized vitals
| Network: 5 -> 12 -> 8 -> 4 -> 1 (ReLU + sigmoid)
| Output: { score, risk, confidence, flaggedVitals }
| Action: write score + risk back to PostgreSQL row
| If risk=critical AND confidence > 0.8:
| triggerAlert() -> Resend email to provider
```


System data flow (continued)

```
|
+--3--> [*** VitalLedger / Blockchain ***]
        hash = SHA-256(canonical JSON of vitals)
        contract.recordReading(patientId, readingId,
                                hash, mlScore)
        returns txHash (Polygon transaction receipt)
        txHash stored in PostgreSQL + returned in API response

[Provider Dashboard]
|
+-- Patient detail: shows mlAnomalyScore badge per reading
|
+-- Verify button: GET /api/vitals/:id/verify
|   -> VitalLedger.verifyReading(dbReadingId, currentVitals)
|   -> recalculates SHA-256 from current DB values
|   -> compares against on-chain stored hash
|   -> returns { valid: true/false, txHash }
|
+-- Print Report PDF:
    includes ML risk level + blockchain txHash
    per row in the 7-day vitals history table
```

Security Note:

Blockchain writes are fire-and-forget — a failure returns txHash="0x0000" and does NOT prevent vital storage in PostgreSQL. SERVER_WALLET_PK must be stored in a secrets manager (Replit Secrets, AWS Secrets Manager) — NEVER committed to git. For production use a Hardware Security Module (HSM) or managed signing service such as AWS KMS or Azure Key Vault.

